



THE LONDON SCHOOL
OF ECONOMICS AND
POLITICAL SCIENCE ■

September 2026 Exam

EC400

Dynamic Programming and Differential Equations

Suitable for all candidates

Instructions to candidates

This paper contains 10 questions. Answer all questions.

Answers should be justified by showing work. Please write your answers in dark ink (black or blue) only.

Only answers written in an answer booklet will be sent for marking.

Time Allowed	Reading Time:	<i>None</i>
	Writing Time:	<i>90 minutes</i>

You are supplied with:	<i>No additional materials</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are not allowed in this exam</i>
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If during this exam you feel you need to make additional assumptions in order to proceed, then please do so by stating your assumptions clearly.

EC400: Exam

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Consider the “pie-eating problem”: each period t , a household decides what fraction of the remaining pie of size B_t to consume C_t to maximize discounted utility

$$\sum_{t=0}^T \beta^t \log C_t,$$

where $T < \infty$ unless stated otherwise below. The law of motion for the size of the pie is given by

$$B_{t+1} = (1 - \delta)(B_t - C_t),$$

where $\delta \in (0, 1)$ is the spoilage rate of the pie and $B_t \geq 0$ in all periods. The initial value B_0 is given.

1. (10 marks) Write down the Lagrangian and take the first-order conditions.
2. (10 marks) Derive the intertemporal optimality condition (Euler equation) and the transversality condition.
3. (5 marks) What are the state and control variables in this problem?
4. (10 marks) Solve for optimal consumption in closed form as a function of the state variables. How does the optimal consumption share of the pie depend on time?
5. (5 marks) From now on, focus on infinite-horizon case $T = \infty$. What are the state variables in this case? What is the transversality condition?
6. (15 marks) Use the law of motion of the pie and the Euler equation to write down a linear dynamic system for (B_t, C_t) . Find the eigenvalues and the eigenvectors of the corresponding matrix.

7. (12 marks) Draw the phase diagram showing the isolines, the steady state(s), the eigenvectors and typical trajectories. For this task, ignore non-negativity constraints on the variables.
8. (10 marks) Show that only one trajectory in the phase diagram satisfies the transversality condition. Do not use the Blanchard-Kahn principle!
9. (10 marks) Define the discount factor ρ such that $\beta = e^{-\rho}$. Write down the Bellman equation for the problem and then derive the corresponding HJB equation.
10. (13 marks) Use discrete-time finite-horizon solution to guess the functional form of the value function in the HJB equation. Solve for its coefficients.